# Power from batteries and the sun



## IoT · Level 3 · Lesson 4

# Today you will

- Explain why a device's radio uses more battery than anything else.
- Make a battery last longer by sleeping and sending less often.
- Describe how a small solar panel keeps a sensor running.

# Make it last

- Hours = capacity ÷ average current
- The radio is the hungriest part
- Sleep most of the time; send less
- Sun by day, battery by night

# Battery sums

1. Find two or three mAh labels.
2. Plan with 70% of each label.
3. Work out the hours at 100 mA and at 10 mA.
4. Plan a garden sensor: what, how often, which radio.
5. Decide: does it need a solar panel?

**Never open, crush, puncture or heat a battery, and never leave lithium batteries or power banks in hot sun. Keep button batteries away from small children.**

# Let’s discuss

1. Which data around you must be fresh every second, and which could wait an hour?
2. Where at your school would a solar-powered sensor be useful?
3. Why does a coin cell suit Bluetooth LE but not Wi-Fi?

# Check yourself

- A sensor sends every minute over Wi-Fi. Name two ways to make its battery last longer.